

AN AGENTIC AI ARCHITECTURE FOR CROSS-BORDER OPEN FINANCE: A HIGH-LEVEL DESIGN PERSPECTIVE BUILDING ON BIS PROJECTS APERTA AND AGORÁ, WITH AN EU–UK CORRIDOR WORKED EXAMPLE

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Abstract

This paper extends a prior architectural thesis on agentic artificial intelligence for national-scale fraud detection to the cross-border Open Finance layer, where the Bank for International Settlements has, through Projects Aperta and Agorá, demonstrated that the data and value layers of cross-border financial infrastructure can be re-architected on shared, programmable substrates. Both BIS projects achieve foundational proof. Both also name explicitly the work that remains. This paper consolidates the eleven gaps named by BIS into a thirteen-gap framework, adds the participant-economics and AI-governance constraints the BIS work does not address, and proposes a layered, role-taxonomised, cost-bounded agentic AI fabric that closes all thirteen. The fabric is anchored in eight regulatory regimes — FATF, PSD3/PSR, ISO 20022, GDPR, MiCA, DORA, the EU AI Act and parallel UK regimes — as design constraints. An EU–UK Open Finance corridor with an Agorá-pattern settlement leg is walked end-to-end, with explicit gap-to-step mapping. Deployment readiness is addressed through eight structured mechanisms and an illustrative unit-economics model showing the participant case is structurally defensible against the empirical record of forty-two percent enterprise AI abandonment.

Keywords: Agentic AI, Cross-border Open Finance, BIS Project Aperta and Agorá, Bounded-autonomy architecture, Participant economics

1. INTRODUCTION

Cross-border payment and data flows now sit at the center of the global financial reform agenda. The G20 Roadmap for Enhancing Cross-Border Payments, adopted in 2020 and refreshed annually since, defines four persistent failure modes that the international financial system has yet to resolve at scale: high costs, slow speeds, limited access, and insufficient transparency [1]. In parallel, the rise of Open Banking in jurisdictions ranging from the United Kingdom and the European Union to Brazil, India, Saudi Arabia, and Bahrain has shifted the policy conversation from payment rails alone to the broader question of cross-border interoperability for financial data and value combined [2][3]. The cumulative result is an architectural moment: the world's central banks, payment system operators, and major financial institutions are converging on the proposition that the next generation of cross-border infrastructure must support data and value flows on shared, programmable substrates that preserve jurisdictional autonomy while enabling end-to-end interoperability.

Two recent initiatives by the Bank for International Settlements crystallize this convergence. Project Aperta, launched by the BIS Innovation Hub, builds a prototype multilateral cross-border interoperability network that connects domestic Open Finance infrastructures of different

jurisdictions, enabling consumer-consented, encrypted financial data to be shared via APIs across borders [4]. Project Agorá, a public-private collaboration convened by the BIS and the Institute of International Finance, has delivered a prototype unified ledger that integrates tokenised central bank reserves with tokenised commercial bank deposits across seven major jurisdictions, demonstrating atomic, multi-currency, around-the-clock settlement of wholesale cross-border payments [5][6]. The two projects together bookend the architectural question: Aperta on the data layer of cross-border Open Finance, Agorá on the value layer of cross-border wholesale settlement. Both have moved decisively beyond conceptual whitepaper to working prototype with named participating institutions and tested workflows.

These are landmark achievements. They are also, by the projects' own admission, foundations rather than finished platforms. The Project Agorá final report, published in May 2026, identifies six explicit areas for future exploration: more coordinated anti-money laundering, sanctions and anti-fraud approaches; liquidity-saving mechanisms; strengthened cybersecurity and operational resilience; advanced interoperability and compatibility requirements; governance, rules and oversight encompassing participation, settlement finality, data governance and risk management; and the technical, operational and contractual definition of settlement finality across legal frameworks [5]. The BIS body of work on Open Finance, including the analytical foundations supporting Project Aperta, identifies parallel gaps in cross-border data portability, cybersecurity investment, jurisdictional heterogeneity in risk preferences and data-sharing norms, and the continued maturation of Open Finance frameworks across implementing jurisdictions [3][4]. Closing these gaps is the agenda this paper addresses.

A prior paper by the author proposed an agentic AI architectural fabric for national-scale fraud detection in digital payments ecosystems, anchored on the NIBSS National Payment Stack in Nigeria under the policy direction set by the Central Bank of Nigeria's January 2025 fraud-debit directive [7]. That paper argued that the workflow bottlenecks of first-generation national fraud platforms — human-in-the-loop investigation, batch retraining latency, and weak cross-institution coordination — are structurally resolvable through a layered architecture of bounded-autonomy agents with persistent memory and observable handoffs. The present paper extends that thesis to the cross-border Open Finance layer. The structural limitations multiply when payment and data flows cross jurisdictional boundaries: consent portability and revocation latency, multi-regulator liability allocation, schema and observability mismatches, and the participant-economics problem of getting commercial banks, payment service providers and other regulated institutions to commit operational capacity to a corridor whose returns are network-dependent. The agentic fabric proposition, refined for this domain, addresses each.

The contribution of this paper is fourfold. First, it surfaces and consolidates the eleven specific gaps that the BIS itself identifies as remaining work after Aperta and Agorá. Second, it maps the regulatory and standards scaffolding — FATF, PSD3 and the Payment Services Regulation, ISO 20022 CBPR+, GDPR, MiCA, DORA, the EU AI Act, and parallel United Kingdom regimes — within which any cross-border agentic architecture must operate. Third, it proposes a layered agentic AI fabric for cross-border Open Finance, with explicit gap-to-design-response mapping, and treats participant economics and institutional protection as first-class architectural concerns rather than afterthoughts. Fourth, it walks an end-to-end worked example through an EU–UK Open Finance corridor with an Agorá-pattern settlement leg, illustrating the fabric in operation.

The paper closes with risks, open questions, and the connection back to the national-layer paper that the present work extends.

2. BACKGROUND AND RELATED WORK

2.1. The cross-border Open Finance landscape in 2026

By 2024, 95 jurisdictions had introduced some form of Open Banking or Open Finance policy framework, with motivations ranging from competition and innovation to financial inclusion and consumer protection [8]. The implementation paths diverge along three broad lines. Regulatory-driven prescriptive approaches mandate banks to share customer-permissioned data through unified API standards and require third parties to register with supervisors, as exemplified by the European Union's Second Payment Services Directive (PSD2) and the United Kingdom's Open Banking framework. Facilitative approaches, used by Singapore, Hong Kong and others, publish technical playbooks and roadmaps while leaving adoption speed and scope to the market. Market-driven approaches, historically characteristic of the United States, evolve through bilateral data-sharing arrangements and private aggregation services, although the Consumer Financial Protection Bureau's Personal Financial Data Rights rule under Section 1033 of the Dodd-Frank Act signals a transition toward a more regulated model [8]. Each approach can be implemented through either centralised or decentralised technical architectures, with India and Türkiye representing the centralised path and Europe's PSD2 implementation and Brazil's Open Finance model representing the decentralised one.

Adoption rates illustrate the heterogeneity. Korea leads with thirty-six million subscribers and one hundred ninety-four million registered accounts as of December 2023 [8]. Brazil records the highest per-capita intensity at fifty-five API calls per user per month as of March 2025 [8]. The United Kingdom and the United States are catching up in user counts and API call volume, while India and Türkiye show rapid growth from lower bases. By December 2024, more than four thousand five hundred third-party providers were operating in retail financial services across the European Union and the United Kingdom, with the majority operating cross-border under European Union passporting rules [8]. The cross-border dimension is therefore not theoretical but already operational at scale within the European Economic Area, and the technical and regulatory architecture supporting it sets the baseline conditions for any new cross-border Open Finance proposition.

2.2. BIS Project Aperta: the data layer

The BIS Innovation Hub launched Project Aperta to address the absence of harmonised cross-border data portability infrastructure for Open Finance. Aperta builds a prototype multilateral cross-border interoperability network connecting the Open Finance infrastructures of different jurisdictions, with initial focus on trade finance use cases such as letters of credit, trade credit insurance and supply chain financing for small and medium enterprises [4]. The prototype enables licensed third-party providers, including banks and fintechs, to exchange consumer-consented, encrypted financial data across jurisdictional boundaries via standardised application programming interfaces. The project explicitly seeks to avoid fragmentation in scope, standards and solutions, which BIS identifies as the principal risk created by the bilateral arrangements that have emerged in some regions [4]. The accompanying BIS analytical work concludes that

realising the promise of Open Finance hinges on continued refinement of frameworks over time and on cross-jurisdictional learning between authorities, while flagging cybersecurity investment costs and heterogeneity in jurisdictional risk preferences as persistent constraints on scaling [3].

2.3. BIS Project Agorá: the value layer

Project Agorá, convened by the BIS and the Institute of International Finance, addresses the parallel question on the wholesale settlement layer. The prototype, whose final report was published in May 2026, brings together seven central banks representing the major convertible currencies — the Banque de France for the Eurosystem, the Bank of Japan, the Bank of Korea, the Bank of Mexico, the Swiss National Bank, the Federal Reserve Bank of New York, and the Bank of England — alongside more than forty regulated financial institutions including JPMorgan Chase, HSBC, Deutsche Bank, Mastercard, Visa, Swift and UBS [5]. The architecture is a two-layer design. A unifying layer records tokenised commercial bank deposits accessible to all participants. A jurisdictional layer hosts independent ledgers where tokenised central bank reserves are recorded under the autonomy and regulatory control of each participating central bank. The design preserves correspondent banking as the backbone of cross-border payments while embedding programmability, atomic multi-currency settlement, around-the-clock operation, parallel compliance controls, and privacy preservation at both token and transaction levels [5]. Atomic settlement, in which all balance updates occur or none at all, eliminates settlement risk and is demonstrated as achievable across all seven participating jurisdictions. The legal analysis confirms that tokenisation as implemented does not alter the legal characterisation of the underlying balances, allowing development within existing legal and regulatory frameworks rather than requiring legislative change [5].

2.4. The prior generation of cross-border experiments

Project Agorá explicitly builds on a prior generation of BIS Innovation Hub experiments, including Project Helvetia and Project Jura, which examined payment settlement using wholesale tokenised central bank reserves on national and cross-currency bases respectively, and on the industry-led Regulated Liability Network exploring tokenised commercial bank deposits, central bank reserves and stablecoins on a shared ledger [5]. The Eurosystem's exploratory work on distributed ledger technology for wholesale central bank money settlement, published in 2025, confirmed strong market interest in using DLT for wholesale transactions and underscored the importance of central bank reserves as a safe settlement asset for DLT-based transactions [9]. A parallel experiment, the multilateral Central Bank Digital Currency Bridge known as mBridge, operated by the central banks of Thailand, the United Arab Emirates, the People's Bank of China Digital Currency Institute and the Hong Kong Monetary Authority, processed four thousand eight hundred sixty-eight cross-border payment transactions totalling approximately four hundred seventy-seven point eight billion in equivalent value by the end of 2025 [10]. Hybrid CBDC architectures, illustrated by jurisdictions implementing both account-based and value-based components selected by scenario, demonstrate the operational viability of scenario-driven architectural choices in payment systems carrying retail and wholesale volumes simultaneously [11].

2.5. The gap addressed by this paper

Aperta proves cross-border Open Finance data interoperability at prototype scale. Agorá proves cross-border tokenised value settlement at prototype scale. Both reports name explicit areas for further work, examined in detail in Section 3. Neither project, by design, addresses how the operational workflows of cross-border Open Finance — onboarding, consent management, fraud detection, anti-money laundering screening, sanctions compliance, dispute resolution, settlement reconciliation, and continuous adaptation to new threat patterns — are coordinated end-to-end across jurisdictions, multiple regulators and heterogeneous institutional participants. The Financial Conduct Authority's October 2025 research note on Open Banking and Open Finance in the United Kingdom acknowledges that the divergent implementation of PSD2 across European Union member states has hindered cross-border interoperability and reduced the effectiveness of competition [2]. The workflow coordination problem is therefore not solved by APIs alone or by tokenised ledgers alone. It requires an active orchestration layer. This paper proposes that layer.

3. STRUCTURAL LIMITATIONS AND THE GAPS IDENTIFIED BY BIS

The two BIS prototypes have moved cross-border Open Finance and cross-border tokenised settlement from concept to working architecture. Both reports also state, with unusual clarity for the genre, what remains undone. This section consolidates the eleven explicit gaps the BIS itself identifies, organises them into three architectural layers, and adds two further limitations that the present paper identifies as structural constraints on adoption at scale. The agentic AI fabric proposed in Section 5 is mapped against each of the resulting thirteen items.

3.1. Gaps explicitly named by Project Agorá

The Project Agorá final report names six areas for future exploration, each accompanied by a concrete operational rationale [5].

The first is more coordinated anti-money laundering, countering the financing of terrorism, sanctions screening and anti-fraud approaches. The prototype runs each participating institution's compliance controls independently and confidentially in parallel, ensures that all required validations complete prior to settlement, and includes functionality that would support enhanced financial crime information-sharing in future iterations. The current design therefore demonstrates parallelism but stops short of coordination. False positive rates in compliance screening, inconsistent data quality across institutions, and limited transparency on transaction status remain unaddressed at the platform layer.

The second is liquidity-saving mechanisms. The prototype scope explicitly excluded these mechanisms, reflecting a deliberate prioritisation of foundational architecture over full optimisation. Cross-border wholesale flows in production environments routinely encounter intra-day liquidity constraints that drive sequencing decisions, queue management requirements and netting arrangements. The absence of these mechanisms from the prototype is therefore not a defect but a deferred design problem of considerable operational consequence.

The third is strengthening cybersecurity posture and operational resilience. The prototype assumptions paragraph states explicitly that production-grade capabilities including cybersecurity, scalability, throughput and latency were not prioritised. For any subsequent platform aspiring to production deployment in jurisdictions subject to the European Union's

Digital Operational Resilience Act, the United Kingdom's Operational Resilience policy framework, or equivalent regimes elsewhere, this is the single largest implementation gap separating the prototype from a regulated financial market infrastructure.

The fourth is advancing interoperability and compatibility requirements. The prototype validates atomic settlement across seven participating jurisdictions but does not yet specify the technical, operational and contractual requirements for onboarding additional currencies, asset classes or institutional types. Foreign exchange integration and tokenised asset classes beyond commercial bank deposits and central bank reserves were placed out of scope for the prototype.

The fifth is governance, rules and oversight covering participation, settlement finality, data governance and risk management. The legal analysis confirms achievability of settlement finality across all seven participating jurisdictions, but defines the operational and contractual requirements as a separate workstream. Governance of participant eligibility, fee structures, dispute resolution mechanisms and rule changes is identified as essential to safe and scalable deployment while preserving jurisdictional autonomy.

The sixth is the technical, operational and contractual definition of settlement finality in alignment with the legal frameworks of participating jurisdictions. Settlement finality is the legal point at which a transaction becomes irrevocable and unconditional, and its technical implementation in a programmable shared-ledger environment under multiple legal regimes is a non-trivial engineering and legal exercise.

3.2. Gaps explicitly named by Project Aperta and the BIS Open Finance evidence base

The analytical work supporting Project Aperta and the BIS Papers No 168 study of international Open Finance experience name five further gaps [3][4][8].

The seventh is cross-border data portability without fragmentation in scope, standards and solutions. Bilateral arrangements emerging in some regions risk fragmentation that reduces interoperability and scalability while increasing overall complexity. The Aperta prototype addresses this for trade finance use cases at pilot scale; the generalisation to retail Open Finance flows across more than two jurisdictions remains a development agenda.

The eighth is cybersecurity investment and the operational costs of secure data sharing systems. The BIS Open Finance evidence base acknowledges that the resource burden of maintaining secure cross-border data exchange falls disproportionately on smaller participants, creating an adoption barrier that asymmetrically affects new entrants and emerging-market institutions.

The ninth is jurisdictional heterogeneity in risk aversion and data-sharing preferences. Authorities and institutions in different jurisdictions exhibit different tolerances for data exposure, different legal bases for consent, different consumer protection postures, and different cultural norms around financial data. The Financial Conduct Authority's October 2025 research note on Open Banking and Open Finance in the United Kingdom acknowledges that the divergent implementation of PSD2 across European Union member states has hindered cross-border interoperability and reduced the effectiveness of competition [2].

The tenth is the continued refinement of Open Banking and Open Finance frameworks over time. The BIS evidence base treats Open Finance as an evolving policy domain rather than a settled

architecture, with implementation still in early days in many jurisdictions and policy frameworks adapting in response to experience.

The eleventh is cross-jurisdictional learning between authorities. The BIS report observes that authorities continue to learn from each other's experience and frames this as an ongoing institutional capacity-building agenda rather than a problem with a defined solution.

3.3. Two further structural limitations identified by this paper

Beyond the eleven explicit BIS gaps, two limitations recur across the operational literature, in the FCA research note and in industry analysis, but receive limited explicit treatment in either Aperta or Agorá. The present paper treats them as first-class architectural concerns.

The twelfth is participant adoption friction. Cross-border infrastructure exhibits network effects: the value of participation to any individual financial institution depends on the number and quality of other participating institutions. Early participants therefore face an adoption risk that late participants do not. Aperta and Agorá assemble their participant pools through BIS and Institute of International Finance convening authority, which is not a mechanism reproducible at scale in production deployments. The economic case for participation by commercial banks, payment service providers and other regulated institutions — including the revenue lines unlocked, the costs reduced, the strategic optionality gained, and the defensive value of avoiding non-participation — must be made explicitly and is rarely articulated in architectural documents. Section 6 of this paper addresses this directly.

The thirteenth is regulatory acceptance heterogeneity for artificial intelligence components. Any cross-border architecture deploying machine learning or agentic AI for compliance, fraud detection or workflow orchestration must navigate the European Union Artificial Intelligence Act, the United Kingdom's pro-innovation regulatory approach to AI, sectoral guidance from supervisors such as the Bank of England Prudential Regulation Authority and the European Banking Authority, and parallel regimes elsewhere [12][13]. The AI Act's risk-tiered approach classifies many financial-sector AI applications as high-risk, triggering documentation, transparency, human oversight, accuracy, robustness and cybersecurity obligations. Aperta and Agorá do not address the AI governance layer, which is appropriate to their scope, but any agentic AI fabric proposition must absorb these obligations as design constraints from the outset.

3.4. Synthesis

The thirteen items above are not independent. They cluster into three interacting architectural layers. The coordination layer comprises gaps one, seven, ten and eleven: the workflow coordination problem across institutions and jurisdictions, both for data sharing and for compliance processing. The resilience and integrity layer comprises gaps three, six, eight and thirteen: cybersecurity, operational resilience, settlement finality and AI governance, all of which must be present and demonstrable for any platform to operate under DORA-equivalent regimes. The economic and adoption layer comprises gaps two, four, five, nine and twelve: liquidity management, expansion path, governance, jurisdictional heterogeneity and the participant adoption case. The agentic AI fabric proposed in Section 5 addresses items concentrated in the coordination layer most directly, while contributing to the resilience and integrity layer through observability and bounded autonomy, and to the economic and adoption layer through the participant economics framework developed in Section 6.

The remainder of this paper proceeds in that order. Section 4 sets out the regulatory and standards scaffolding within which any cross-border agentic architecture must operate, mapping each of the thirteen items to the specific regulatory levers that constrain or enable its resolution. Section 5 proposes the agentic AI fabric and maps each agent role to the gaps it closes. Section 6 develops the participant economics and institutional protection framework. Section 7 walks an EU–UK Open Finance corridor with an Agora-pattern settlement leg end-to-end. Section 8 surveys risks and open questions. Section 9 concludes.

4. THE REGULATORY AND STANDARDS SCAFFOLDING

Any architecture for cross-border Open Finance operates inside an unusually dense regulatory perimeter. Eight regimes converge on the same operational space: the Financial Action Task Force standards on anti-money laundering and counter-terrorism financing; the European Union's Third Payment Services Directive and accompanying Payment Services Regulation; the International Organization for Standardization 20022 messaging standard and its Cross-Border Payments and Reporting Plus profile; the General Data Protection Regulation and parallel United Kingdom data protection regime; the Markets in Crypto-Assets Regulation; the Digital Operational Resilience Act; the European Union Artificial Intelligence Act; and the United Kingdom's principles-based pro-innovation approach to AI regulation. This section treats these regimes not as compliance overhead but as design constraints that shape the agentic fabric's architecture. The mapping is deliberate: every architectural decision in Section 5 is anchored in one or more of the regulatory levers introduced here, and every closure of one of the thirteen gaps from Section 3 must be defensible against the relevant regime.

4.1. The compliance and integrity regimes: FATF, PSD3 and PSR, MiCA

The Financial Action Task Force standards define the global baseline for anti-money laundering, counter-terrorism financing and sanctions screening, including the so-called Travel Rule which requires originator and beneficiary information to accompany cross-border transfers above defined thresholds. For an agentic fabric operating across jurisdictions, FATF compliance is not a single check but a set of distributed checks that must be coordinated without compromising the privacy controls that the architecture also commits to preserving. Project Agora's prototype runs each institution's screening independently and in parallel, completing all required validations before settlement [5]. The fabric proposed in Section 5 extends this with an explicit coordination agent operating under a bounded-autonomy mandate: it does not share screening results across institutions where consent or legal basis is absent, but it does share match metadata, false-positive feedback signals, and confirmed-typology indicators that enable each institution's local screening to improve over time. This addresses gap one (coordinated AML/CFT/sanctions/anti-fraud approaches) without breaching the privacy controls that gap nine (jurisdictional heterogeneity in data-sharing preferences) requires.

The European Union's Third Payment Services Directive and accompanying Payment Services Regulation, replacing PSD2 from 2026 onward, reallocate liability for authorised push payment fraud, expand strong customer authentication exemption rules, introduce mandatory refund obligations under defined circumstances, and impose new transparency obligations on payment service providers regarding fraud rates and fee structures [17]. The PSR is directly applicable across European Union member states without national transposition, addressing the divergent-

implementation problem that PSD2 created [18]. For a cross-border Open Finance corridor, three PSD3/PSR design constraints are decisive. First, the fabric must support real-time fraud signalling between institutions involved in a payment chain to allocate liability correctly when fraud is discovered post-settlement. Second, the consent management layer must implement the PSR's enhanced consent dashboard and revocation requirements with sub-day latency. Third, the fraud-rate transparency obligation creates a structured incentive for the fabric to maintain low aggregate fraud rates, since participating institutions must publish them. The agentic fabric's bounded-autonomy fraud agents, operating with persistent memory across the corridor, are calibrated for this incentive structure.

The Markets in Crypto-Assets Regulation establishes the European Union licensing regime for crypto-asset service providers and issuers of asset-referenced tokens and electronic money tokens. The relevance for the present paper is bounded but specific: where the agentic fabric integrates with an Agorá-pattern settlement leg involving tokenised commercial bank deposits or tokenised central bank reserves, the legal characterisation of these instruments matters for the regulatory perimeter under which the fabric operates. Project Agorá's legal analysis confirmed that tokenisation as implemented in the prototype does not alter the legal characterisation of the underlying balances or the nature of the obligations represented [5]. Tokenised commercial bank deposits remain commercial bank money; tokenised central bank reserves remain central bank money. MiCA does not therefore apply to the Agorá-pattern settlement leg of the corridor, since these are not crypto-assets within the MiCA definition. The fabric architecture must, however, remain capable of distinguishing the regulatory classification of each token type it handles, since corridors extending beyond Agorá-style arrangements may include stablecoins or other instruments that fall within MiCA's scope.

4.2. The data and messaging regimes: ISO 20022, GDPR and UK data protection

The International Organization for Standardization 20022 messaging standard, and specifically its Cross-Border Payments and Reporting Plus profile, defines the structured payload format for cross-border payments messages. The Bank for International Settlements Committee on Payments and Market Infrastructures has driven CBPR+ adoption as part of the G20 Roadmap for Enhancing Cross-Border Payments [1]. Project Agorá's prototype incorporates ISO 20022 CBPR+ as well as Legal Entity Identifiers and Confirmation of Payee best practices [5]. For the agentic fabric, ISO 20022 is the lingua franca: every agent operates on structured, validated payloads with deterministic semantics, eliminating the schema ambiguity that has historically been one of the most expensive sources of cross-border payment failure. The fabric does not invent its own data model; it operates on the existing standard, with agents acting as policy and orchestration layers above it.

The General Data Protection Regulation defines the European Union data protection regime, with the United Kingdom General Data Protection Regulation and the Data Protection Act 2018 establishing the parallel United Kingdom regime [19]. Both regimes converge on the same set of design constraints relevant to the fabric: lawful basis for processing, purpose limitation, data minimisation, accuracy, storage limitation, integrity and confidentiality, and accountability. The Data (Use and Access) Act, advancing through the United Kingdom legislative process as of 2025, provides the statutory foundation for the United Kingdom's smart data and Open Finance frameworks [2]. For the fabric, three design implications follow. First, every agent operates under

a documented lawful basis registered in the consent management layer. Second, persistent agent memory is scoped to the purpose for which consent was given, with explicit retention limits and right-to-erasure mechanics. Third, cross-border data transfers between the United Kingdom and the European Union remain governed by the adequacy decision granted by the European Commission, with the agentic fabric architected to fall back to standard contractual clauses or equivalent mechanisms should the adequacy decision lapse.

4.3. The operational resilience and AI governance regimes: DORA, EU AI Act, UK AI approach

The Digital Operational Resilience Act establishes binding requirements for information and communication technology risk management, incident reporting, digital operational resilience testing, third-party ICT risk management, and information-sharing arrangements among regulated financial entities [20]. DORA applies across the European Union financial sector and extends to ICT third-party service providers including cloud providers, software vendors and any party providing ICT services to a financial entity. For the agentic fabric, DORA is the regulatory regime that most directly addresses Project Agorá's third gap (strengthening cybersecurity posture and operational resilience). The fabric's design must include continuous monitoring, classified incident reporting workflows, threat-led penetration testing capability, third-party ICT risk register integration, and operational continuity provisions. The United Kingdom's operational resilience framework, set out in the Bank of England Prudential Regulation Authority Supervisory Statement on operational resilience and the Financial Conduct Authority's equivalent guidance, imposes parallel requirements with slightly different formulation but comparable substantive obligations.

The European Union Artificial Intelligence Act establishes a risk-tiered regulatory framework for AI systems, with high-risk classifications attaching to AI used in critical financial-sector applications including creditworthiness assessment and certain biometric uses [12]. The AI Act imposes obligations on high-risk AI systems covering risk management, data governance, technical documentation, record-keeping, transparency, human oversight, accuracy, robustness and cybersecurity. The Qatar Central Bank AI Guidelines, the European Banking Authority guidance on machine learning in internal models, and parallel supervisory statements from other authorities establish sector-specific overlays [13][21]. The United Kingdom's pro-innovation approach to AI regulation, while less prescriptive, converges on similar substantive expectations through cross-cutting principles applied by sectoral regulators including the Financial Conduct Authority, the Bank of England and the Information Commissioner's Office. For the fabric, the AI Act and parallel regimes are not optional add-ons. Every agent in the architecture must be classified by AI Act risk tier, every high-risk agent must comply with the full documentation and oversight obligations, and every cross-border deployment must navigate the consistency questions between the European Union regime and the United Kingdom equivalent. Section 5's architectural design takes this as a starting constraint, not a downstream consideration.

4.4. Synthesis: the eight regimes as enablers, not obstacles

The eight regimes converge on five operational design imperatives that the agentic fabric must satisfy by construction rather than by retrofit. The fabric must provide auditable observability of every agent action sufficient to support DORA incident reporting and AI Act human oversight obligations. It must operate within explicit, registered legal bases for each processing activity,

satisfying GDPR and UK GDPR. It must support structured, validated message semantics, satisfying ISO 20022 CBPR+. It must coordinate compliance processing across institutions without breaching jurisdictional data-sharing constraints, satisfying FATF, PSD3/PSR and Project Agora's gap one simultaneously. And it must operate under bounded autonomy with cost-controlled, role-taxonomised agents, satisfying both the AI Act's human oversight requirements and the operational cost constraints that the next sections develop. The architecture proposed in Section 5 is built around these five imperatives. Each closes one or more of the thirteen gaps named in Section 3, and each is anchored in one or more of the eight regulatory regimes named here.

5. THE AGENTIC AI FABRIC FOR CROSS-BORDER OPEN FINANCE

The architecture proposed in this section is shaped by three commitments. First, every agent operates under bounded autonomy, role-taxonomised, with explicit termination conditions and observable handoffs — not because this is aesthetically preferable but because it is the only configuration in which the architecture's operating cost is predictable and recoverable through participant economics. Second, the fabric is scenario-driven: agentic autonomy is deployed at the layers where it earns its cost, and traditional rails operate at the layers where atomicity, finality and regulatory determinism take precedence. Third, the architecture treats the five regulatory design imperatives identified in Section 4 — observability, registered legal basis, structured semantics, coordinated compliance under jurisdictional autonomy, and bounded-autonomy AI governance — as construction constraints, not as downstream compliance work. The result is a layered, scenario-driven, cost-bounded agentic fabric whose deployment economics are defensible from inception.

5.1. First design principle: cost discipline as architectural law

The contemporary empirical record on enterprise AI deployment is unambiguous. The Deloitte 2025 AI Adoption Survey records that forty-two percent of enterprises abandoned most of their AI initiatives in 2025, more than double the seventeen percent abandonment rate of the previous year [14]. The Massachusetts Institute of Technology 2025 study on generative AI pilots found that eighty-eight percent never reach production deployment [22]. Gartner projects that more than forty percent of agentic AI projects will be cancelled by the end of 2027, citing escalating costs, unclear business value and inadequate risk controls [15]. The FinOps Foundation's 2026 State of FinOps Report identifies AI and data platforms as the fastest-growing new category of enterprise spend, driven by what the report names "token-maxing": the organisational default of routing every task to the most capable and expensive model with no governance, no routing logic and no cost visibility [16]. The Mavvrik and Benchmarkit study of three hundred seventy-two enterprises found that only fifteen percent forecast their AI costs within ten percent of actual spend [23]. A March 2026 audit of thirty engineering teams running agentic AI in production identified a twenty-times spread between the tenth and ninetieth percentile of per-developer cost, with one growth-stage company recording an eighty-seven thousand United States dollar agent bill in a single month [24]. Industry response in 2026 has converged on a new discipline known as FinOps for AI, with vendors releasing dedicated cost-control modules including Portal26's Agentic Token Controls, which let administrators set token budgets per agent, throttle agents approaching caps and terminate agents that breach them [25].

These failures are not failures of agentic AI as such. They are failures of unbounded, untaxonomised, unmonitored AI deployment. The architecture proposed here is the structural opposite. Five design choices, taken at the architectural level rather than as downstream optimisations, foreclose by construction each named failure mode.

First, the fabric is organised as a role-taxonomised population of specialised agents, not a single general-purpose agent that handles every task. The fraud detection agent does not write code. The consent coordination agent does not run anti-money laundering screening. The settlement reconciliation agent does not perform sanctions matching. Each agent's mandate is defined narrowly enough that the most expensive model class is not required for the great majority of its work.

Second, the fabric implements tiered model routing as a protocol-layer requirement, not a participant-side optimisation. Routine work — schema validation, structured data transformation, deterministic rule application, the great majority of compliance checks — runs on smaller, cheaper models or on rules engines outside the AI envelope entirely. Frontier model capability is escalated to only when bounded-autonomy policy demands it, typically for ambiguous edge cases, novel typology detection or supervised investigation. The LeanOps audit of thirty teams that adopted comparable routing patterns recorded cost reductions of fifty-five to seventy-five percent within thirty days of adoption [24]. The architecture specifies this routing as a fabric-wide invariant, not a recommendation.

Third, every agent operates under explicit token and step budgets registered at instantiation. Agents approaching budget caps are throttled. Agents exceeding budget caps are terminated, with the partial work checkpointed for human handoff. This is the Portal26 pattern operationalised at the protocol layer rather than as an after-market control [25].

Fourth, persistent agent memory is bounded by purpose, jurisdiction and consent. Memory is not a single ever-growing context window but a structured store with explicit retention limits, sliding-window pruning for working memory, and consented persistence for institutional learning. Context-window expansion is the single largest source of agent cost runaway in production [24]; the architecture forecloses it by design.

Fifth, the fabric is deployed in a federated configuration. Compliance-processing agents run inside each financial institution's own perimeter on infrastructure the institution already operates. Data-sharing and orchestration agents run inside the regulated interconnection layer. Settlement-related agents interact with the Agorá-pattern ledger through pre-validated payloads. Only orchestration metadata and consented data crosses institutional boundaries. The cost surface is therefore distributed across participants by design, not concentrated in a single tenant's bill by accident.

5.2. The architectural layers

Figure 1. Layered Architecture of the Cross-Border Open Finance Agentic Fabric



Figure 1. Layered architecture of the cross-border Open Finance agentic fabric: six interacting layers with the agentic orchestration layer cutting vertically through the consumer, regulated participant, cross-border data interconnection, wholesale settlement and supervisory observability layers.

The fabric is organised as six interacting layers, illustrated in Figure 1. The consumer and small-business layer holds the data subjects and beneficiaries of cross-border Open Finance services, with consent dashboards exposing the current state of all authorisations and the right to revoke. The regulated participant layer comprises the financial institutions, microfinance institutions, non-bank financial institutions, payment service providers and other licensed entities that interact with the data subjects and that operate the compliance-processing agents on their own infrastructure. The cross-border data interconnection layer implements the Aperta-pattern interoperability network, exchanging consumer-consented, encrypted financial data via standardised application programming interfaces between licensed third-party providers across jurisdictions. The wholesale settlement layer implements the Agorá-pattern two-layer architecture, with a unifying ledger for tokenised commercial bank deposits and jurisdictional ledgers for tokenised central bank reserves. The supervisory observability layer provides each participating authority with the regulator-facing view of the activity under its jurisdiction, with auditable logs of every agent action sufficient to support DORA incident reporting and AI Act human oversight. The agentic orchestration layer cuts vertically through all the others, hosting the role-taxonomised population of agents that coordinate workflows end-to-end while respecting the boundaries the other layers define.

The layering is not decorative. Each layer corresponds to a regulatory regime under which it predominantly operates: consumer layer under GDPR and UK GDPR; participant layer under PSD3/PSR, FATF and sectoral supervision; interconnection layer under Aperta-pattern interoperability requirements; settlement layer under the legal frameworks of the participating central banks; observability layer under DORA, the EU AI Act and parallel UK frameworks; orchestration layer under all of the above as the bounded-autonomy AI governance perimeter. An agent operating in one layer cannot exceed the regulatory envelope of that layer without an explicit escalation to a higher-authority agent operating under the appropriate combined authority. This containment is what makes the architecture defensible to supervisors.

5.3. The agent role taxonomy

The architecture defines ten agent roles, organised into three functional families.

The compliance and integrity family comprises four roles. The consent coordination agent manages the lifecycle of every data-sharing authorisation, including PSR-compliant dashboard exposure, sub-day revocation propagation and lawful-basis documentation. The AML and sanctions coordination agent orchestrates the parallel screening processes across participating institutions, shares match metadata and false-positive feedback signals under documented legal basis, and identifies confirmed typology indicators that improve local screening over time. The fraud detection agent operates on transaction and behavioural data within each institution's perimeter, with persistent memory of confirmed typologies and continuous learning loops bounded by purpose and consent. The liability allocation agent tracks fraud events across the payment chain post-settlement, supporting PSD3/PSR refund obligations and producing the audit trail that allocates liability correctly between participants.

The data and value orchestration family comprises three roles. The data portability agent implements the Aperta-pattern data exchange across the interconnection layer, validating payloads against ISO 20022 CBPR+ where applicable and against Open Finance API specifications elsewhere. The settlement orchestration agent interacts with the Agorá-pattern ledger, performing pre-settlement validation, atomic settlement triggering and post-settlement reconciliation. The liquidity coordination agent addresses Project Agorá's second gap by managing intra-day liquidity positions across the corridor, supporting queue management and netting arrangements within the bounded-autonomy envelope.

The supervisory and resilience family comprises three roles. The supervisory observability agent exposes the regulator-facing view of activity under each authority's jurisdiction, producing the audit logs that DORA incident reporting and AI Act human oversight require. The resilience and incident agent monitors the fabric for operational disruptions, classifies incidents under DORA taxonomy, triggers reporting workflows and coordinates resilience testing including threat-led penetration testing exercises. The governance agent manages participant onboarding, eligibility verification, rule change propagation and the dispute resolution workflow.

Each of the ten agents is classified by EU AI Act risk tier at deployment, with the high-risk classifications attracting full documentation, transparency, human oversight, accuracy, robustness and cybersecurity obligations. The role taxonomy is not exhaustive — corridors with specialised use cases may add roles for, for example, trade finance documentation or specific Aperta use cases — but it is sufficient to address the thirteen gaps from Section 3 in the configurations described in subsequent sections.

5.4. Memory substrate and orchestration

Figure 2. Three-Tier Memory Substrate with Consent and Legal-Basis Gating

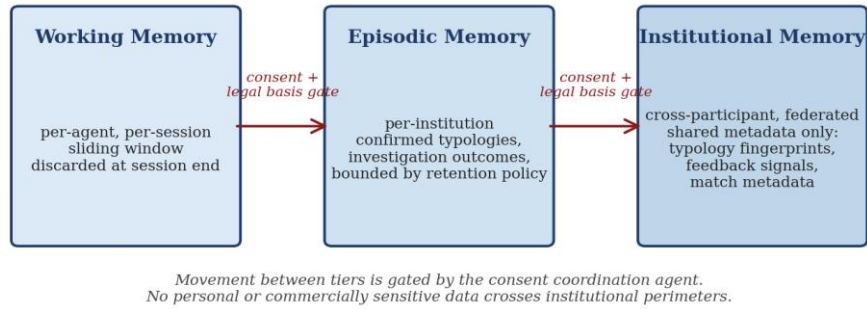


Figure 2. Three-tier memory substrate. Working memory is per-agent and per-session, discarded at session termination. Episodic memory is per-institution, bounded by retention policy. Institutional memory is cross-participant and federated, holding only metadata shared under documented legal basis.

Persistent agent memory in the fabric is organised as a three-tier substrate, illustrated in Figure 2. Working memory is per-agent and per-session, bounded by an explicit sliding window and discarded at session termination. Episodic memory is per-institution, holding the confirmed typologies, investigation outcomes and learning signals that improve the agent's local performance over time, bounded by retention policy under the institution's GDPR/UK GDPR documentation. Institutional memory is cross-participant, federated across the fabric, containing only the metadata explicitly shared under documented legal basis — confirmed typology indicators, fraud-pattern fingerprints, sanctions-match feedback signals — and never the underlying personal or commercially sensitive data from which those metadata are derived. Movement between tiers is gated by consent and legal-basis verification, with the consent coordination agent acting as the gatekeeper.

Orchestration follows a publish-subscribe pattern with explicit handoff semantics. Each agent declares the events it consumes, the events it produces, the maximum autonomy budget it is willing to operate under and the escalation conditions under which it will hand control to a higher-authority agent or to a human supervisor. The orchestration agent maintains the registry of these declarations and routes events accordingly. Handoffs are observable: every transition is logged with sufficient detail to support DORA incident reporting and AI Act human oversight, and the supervisory observability agent exposes this log to each participating authority under its jurisdiction.

5.5. Gap-closure mapping

Figure 3. Gap-Closure Matrix: BIS-Identified Gaps and Paper-Identified Gaps × Architectural Mechanisms

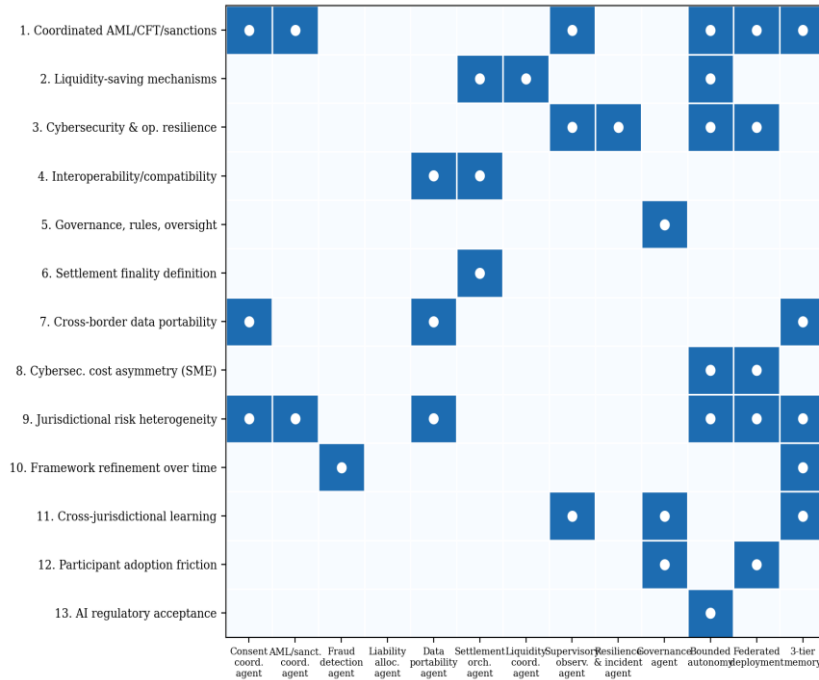


Figure 3. Gap-closure matrix. Rows enumerate the thirteen gaps from Section 3 — the eleven explicitly named by Project Agorá and Project Aperta plus the two paper-identified gaps. Columns enumerate the ten agent roles and the three cross-cutting architectural mechanisms. Filled cells indicate the architectural mechanisms that close each gap.

The architecture as described closes the gaps from Section 3 as follows. The coordination layer gaps (one, seven, ten, eleven — coordinated AML/CFT, cross-border data portability without fragmentation, framework refinement, cross-jurisdictional authority learning) are addressed by the AML/sanctions coordination agent, the data portability agent and the institutional memory tier, all operating under the consent coordination agent's gating. The resilience and integrity layer gaps (three, six, eight, thirteen — cybersecurity/operational resilience, settlement finality definition, cybersecurity investment cost asymmetry, AI governance heterogeneity) are addressed by the supervisory observability agent, the resilience and incident agent, the federated deployment model and the bounded-autonomy/role-taxonomy architecture. The economic and adoption layer gaps (two, four, five, nine, twelve — liquidity-saving mechanisms, interoperability/compatibility expansion, governance, jurisdictional heterogeneity, participant adoption friction) are addressed by the liquidity coordination agent, the governance agent, the federated deployment model and — in the largest share — by the participant economics framework developed in Section 6. The architecture is therefore not a single mechanism that closes all thirteen gaps. It is a layered, role-taxonomised, cost-bounded population of mechanisms whose collective operation closes them, with the economic case for adoption developed as a parallel architectural concern in the next section. Figure 3 summarises the mapping in matrix form.

6. PARTICIPANT ECONOMICS AND INSTITUTIONAL PROTECTION

A cross-border architecture, no matter how elegant, is dead on arrival if the institutions whose participation it requires cannot make the budget case. This section develops that case. It quantifies the addressable fraud loss against which the fabric's value must be measured, frames the participant decision as a cost-benefit comparison rather than an aspirational investment, ports the

consumer protection literature up to the institutional layer, and closes with the structured framework under which a chief financial officer can defend participation to a board.

6.1. The addressable fraud loss as the empirical anchor

The Nasdaq Verafin 2026 Global Financial Crime Report records five hundred seventy-nine point four billion United States dollars lost to bank fraud and scams globally in 2025, an increase of fifty-three point three billion dollars from 2023, with much of the increase attributed to bad actors leveraging artificial intelligence [26]. Independent industry forecasts project that financial institution fraud losses alone, distinct from the consumer-borne losses, will rise from under twenty-five billion dollars in 2025 to more than fifty-five point three billion dollars by 2030, a one hundred fifty percent increase reflecting the structural pressure that AI-enabled fraud places on institutions whose defensive architectures have not kept pace [27]. The London Stock Exchange Group's Risk Intelligence unit projects that global losses from authorised push payment scams alone could reach three hundred thirty-one billion dollars by 2027, with the European Economic Area recording one point one three billion euros in fraudulent transfers in the first half of 2023 of which fifty-seven percent were authorised push payment scams [28]. In the United Kingdom, authorised push payment fraud accounted for four hundred sixty million pounds in losses in 2023, representing forty percent of all payment fraud [28].

These numbers establish the addressable problem space within which the fabric's value must be evaluated. Cross-border flows are a disproportionate share of the loss surface: the wholesale cross-border payment market reached forty-four point one trillion dollars in 2023 and is projected to reach sixty-five trillion dollars by 2030 [29], and a one basis point reduction in cross-border fraud loss rate at that volume releases value that no single participating institution could capture through internal investment alone. Machine learning fraud prevention, where deployed in mature configurations, already prevents an estimated eight billion dollars in losses annually with documented reductions of approximately forty percent against the pre-deployment baseline [29]. The agentic fabric proposed in this paper extends that achievement structurally: by coordinating fraud and AML signal processing across institutions and across jurisdictions through the bounded-autonomy mechanisms described in Section 5, it captures value that institutional-silo deployments cannot.

6.2. The participant decision as a cost-benefit comparison

A chief financial officer evaluating participation in the fabric faces a comparison, not an investment decision. The comparison is between three options. The first is to continue operating with current institutional fraud and compliance architecture and absorb the rising loss rate documented above. The second is to invest in a unilateral artificial intelligence upgrade of the existing institutional architecture, which exposes the institution to the cost-runaway pattern documented in Section 5: a forty-two percent abandonment rate, eighty-five thousand United States dollars in average monthly AI spend with poor cost predictability, and a probability of project cancellation that Gartner places above forty percent for agentic AI projects by end of 2027 [14][15][16]. The third is to participate in the fabric, in which the cost of the AI capability is shared across participants, the architecture's bounded-autonomy design forecloses cost runaway by construction, and the loss reduction benefit accrues to each participant institution proportionally to its transaction volume in the fabric.

The third option is the only one with a defensible cost-benefit ratio under current conditions, and the architecture proposed here makes that defensibility structural rather than aspirational. The cost discipline embedded in Section 5 — role taxonomy preventing token-maxing, tiered model routing reducing per-task cost by fifty-five to seventy-five percent against unbounded baselines [24], explicit token and step budgets per agent, federated deployment distributing the cost surface across participants — converts the AI invoice from an unpredictable line item into a forecastable operational expense. The Mavvrik and Benchmarkit study finding that only fifteen percent of enterprises forecast their AI costs within ten percent of actual [23] applies to unbounded deployments. The bounded-autonomy fabric's cost is forecastable because every agent operates under a registered budget and every escalation is observable.

This is the cost-benefit argument that incentivises participation. A mid-sized European bank processing twenty thousand cross-border transactions per month, with a current addressable fraud loss attributable to those transactions estimated in the low millions of euros annually, can compare three numbers: the loss currently absorbed; the all-in cost of a unilateral AI build with its associated abandonment risk; and the participation fee in the fabric plus the residual loss after the fabric's coordinated detection reduces fraud rate. The fabric is incentive-compatible when the third number is lower than the first two, and the architecture is designed to make this routinely true. The decisive structural fact is that the fabric's coordinated signal-sharing — confirmed typology indicators, false-positive feedback, sanctions match metadata — operates as a positive externality: each participant's contribution improves every other participant's detection rate, and the cumulative loss reduction at the corridor level exceeds what any institution could achieve unilaterally.

6.3. Revenue, cost reduction and strategic optionality

Beyond loss avoidance, three structured value streams support the participation case.

Direct revenue lines. The fabric exposes commercial interfaces that participating institutions can monetise. Interchange and switching arrangements on cross-border account-to-account flows generate per-transaction revenue analogous to existing card scheme economics but at a fraction of card-network cost. Premium application programming interface access to data enrichment, identity verification and behavioural signals creates a subscription revenue line on top of the regulatory data-sharing baseline. Embedded finance partnerships, where the participating institution provides regulated infrastructure for non-financial brands operating in the corridor, generate sponsor-bank fee revenue on a per-account or per-transaction basis. None of these revenue lines depend on the institution displacing existing relationships; they extend the institution's commercial surface into the cross-border corridor.

Cost reduction lines. The fabric reduces operational costs along three axes. Fraud loss absorption is shared across the corridor rather than borne unilaterally by the institution at which the fraud event manifests. Compliance processing costs are reduced through the AML coordination agent's parallel-then-shared architecture, with documented industry baselines of forty percent improvement in straight-through processing rates when coordinated screening replaces sequential institutional silos [29]. Investigation costs are reduced through the persistent memory substrate, which retains confirmed typology indicators that previously required re-investigation at each occurrence. Reconciliation costs are reduced by the Agorá-pattern atomic settlement leg, which

eliminates the post-settlement unwinding workflows that today account for a substantial share of cross-border payment operational expenditure [5].

Strategic optionality. Participation in the fabric secures first-mover position in the corridor's customer base. As cross-border Open Finance adoption matures, customers — both retail and corporate — increasingly expect seamless cross-border experiences, and institutions outside the fabric face customer attrition to participants. The defensive value of avoiding non-participation grows with corridor maturity. Beyond defensive value, the fabric provides early access to product configurations that non-participating institutions cannot offer at all: conditional cross-border payments triggered by Open Finance data events, around-the-clock settlement of corporate treasury flows, and the embedded finance use cases the Agorá legal analysis confirms are within the existing regulatory perimeter [5].

6.4. Network effects and the participant acquisition path

Cross-border infrastructure exhibits the network effects familiar from payment systems generally: the value of participation to any individual institution depends on the number and quality of other participating institutions. The classical network-effects challenge — early participants face adoption risk that late participants do not — is mitigated by three structural choices in the fabric's design. First, capability tiering allows institutions to opt into a subset of the agent population at lower entry cost; an institution can participate in the consent coordination agent and the AML coordination agent without committing to settlement participation. Second, the public-private cost-sharing model, with the Bank for International Settlements and Institute of International Finance convening pattern as the reference, absorbs the core infrastructure development cost on a not-for-profit recovery basis through a permanent foundation or industry utility, with participants paying variable transaction fees tied to use. Third, sandboxed launch in each participating jurisdiction's existing regulatory innovation environment — the Financial Conduct Authority's regulatory sandbox, the European Banking Authority's innovation hub, the Bank of England's New Bank Start-up Unit and the Authority for the Financial Markets' parallel facilities — eliminates the where-do-we-test-this question and accelerates time to first live-value transaction. Section 8 develops these mechanisms in detail.

6.5. Institutional protection: porting the consumer protection literature up

The consumer protection literature in digital finance is mature. The Consultative Group to Assist the Poor's analysis of solutions to protect consumers from fraud sets out a framework of disclosure, redress, complaints handling, fair treatment, predictable liability and protection from unilateral rule changes [30]. The World Bank's Good Practices for Financial Consumer Protection, the Organisation for Economic Co-operation and Development High-Level Principles, and the European Banking Authority Guidelines on complaints handling converge on the same primitives at the consumer-institution interface. The argument advanced in this paper is that the participating institutions themselves face an isomorphic set of protections at the institution-architecture interface, and the fabric must provide them.

Five primitives translate. Predictable liability allocation: the fabric specifies in advance which participant bears which loss in which scenario, with the liability allocation agent producing the audit trail that supports PSD3/PSR refund obligations and that protects participants from unilateral post-hoc reallocation by other participants or by the fabric operator. Audit-grade

observability: every participant has cryptographic-grade access to the actions taken on its behalf by fabric agents, with the supervisory observability layer providing the regulator-facing view to each authority under its jurisdiction. Fair access to redress: the governance agent operates a dispute resolution workflow with documented timelines, escalation paths and arbitration provisions that no individual participant can override. Protection against unilateral rule changes: the governance model requires multi-participant consent for rule changes affecting the architecture's operational parameters, with grandfathering provisions for existing participants and clear notice periods for changes. Protection against vendor lock-in: the fabric's protocol layer is specified as an open reference architecture with no per-participant license fee, and commercial value sits at the application and services layer where competition is permissible. Each of these five primitives mirrors a consumer-protection primitive scaled to the institutional layer. The argument is not metaphorical: institutional participants in a cross-border infrastructure face the same asymmetric-power risks that consumers face in financial services, and the architecture must provide the same structural protections. The fabric does.

7. WORKED EXAMPLE: AN EU–UK OPEN FINANCE CORRIDOR WITH AN AGORÁ-PATTERN SETTLEMENT LEG

The architecture proposed in Section 5, the regulatory scaffolding mapped in Section 4 and the participant economics framed in Section 6 are now operationalised against a single concrete corridor. The European Union and United Kingdom pairing is chosen as the worked example for four reasons. Both jurisdictions are mature Open Banking implementations with the largest active third-party provider populations in the world. Both are subject to substantively comparable regulatory regimes following the European Commission's adequacy decision on UK data protection, while differing in implementation detail in ways that exemplify the cross-border friction problem. Both are participating central bank jurisdictions in Project Agorá, with the Banque de France acting for the Eurosystem and the Bank of England representing the United Kingdom [5]. And both have functioning regulatory innovation environments — the European Banking Authority's innovation hub, the Bank of England's New Bank Start-up Unit, and the Financial Conduct Authority's regulatory sandbox — within which the fabric can be tested at controlled scale before production deployment. This section walks the corridor end-to-end, demonstrates the agentic fabric in operation across a representative transaction, and connects each operational step back to the gap-closure logic from Section 5.

7.1. Corridor configuration

Figure 4. EU-UK Open Finance Corridor Topology
European Union side **United Kingdom side**

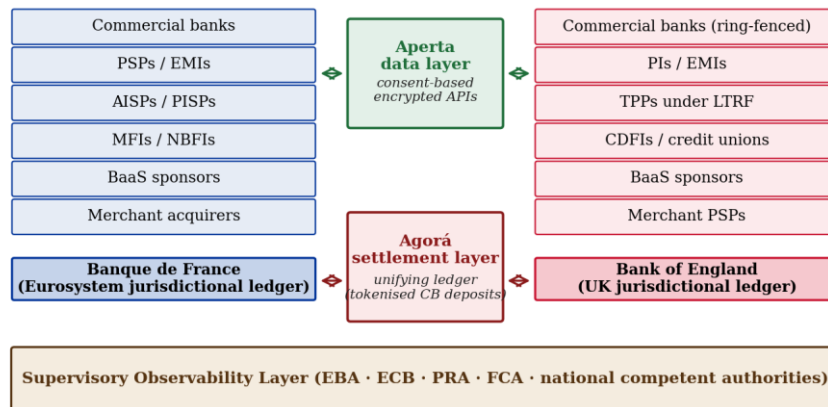


Figure 4. EU–UK Open Finance corridor topology. Participating institutions of seven categories on each side connect through the Aperta-pattern data interconnection layer and the Agorá-pattern settlement layer comprising the unifying ledger and the Banque de France and Bank of England jurisdictional ledgers. The supervisory observability layer exposes parallel regulator-facing views to the EBA, ECB, PRA and FCA.

The corridor configuration involves four categories of participants on each side. On the European Union side, participating institutions include: a representative panel of commercial banks holding tokenised commercial bank deposit issuance authority; payment service providers operating under PSD3 and the Payment Services Regulation, including payment institutions and electronic money institutions; account information service providers and payment initiation service providers operating under the European Union third-party provider regime; microfinance institutions and non-bank financial institutions licensed under national regulators or operating cross-border under European passporting; banking-as-a-service sponsor banks providing regulated infrastructure to embedded finance partners; merchant acquirers and payment facilitators connected to the corridor through the embedded data exchange leg; and the Banque de France operating the Eurosystème jurisdictional ledger for tokenised central bank reserves. On the United Kingdom side, participating institutions include: United Kingdom-incorporated commercial banks and ring-fenced bank subsidiaries holding deposit-taking authority; United Kingdom-authorised payment institutions and electronic money institutions; third-party providers operating under the United Kingdom Open Banking framework now consolidated under the Long-Term Regulatory Framework arising from the Data (Use and Access) Act; community development financial institutions, credit unions and microfinance providers operating under Financial Conduct Authority authorisation; banking-as-a-service sponsor banks providing the regulated layer to United Kingdom embedded finance platforms; merchant payment service providers and card scheme participants integrating Open Banking payments at checkout; and the Bank of England operating the United Kingdom jurisdictional ledger.

The cross-border data interconnection layer implements the Aperta pattern, with consent-based encrypted data flows mediated by the data portability agent across the API perimeter. The wholesale settlement leg implements the Agorá pattern, with the unifying ledger recording tokenised commercial bank deposits accessible to all participants and the two jurisdictional

ledgers recording tokenised central bank reserves under the respective central banks' autonomy and regulatory control.

The supervisory observability layer is dual-headed in the corridor. The European Banking Authority and the European Central Bank receive the regulator-facing view of European Union side activity through the supervisory observability agent operating under the European Union AI Act and DORA. The Prudential Regulation Authority and the Financial Conduct Authority receive the parallel view of United Kingdom side activity through the equivalent agent operating under the United Kingdom operational resilience framework and the United Kingdom approach to AI regulation. National competent authorities for individual member states retain their existing supervisory perimeters where applicable, with the fabric providing the technical means to surface the relevant view to each. Cross-jurisdictional supervisory cooperation, where required, follows the existing memoranda of understanding between the relevant authorities, with the fabric providing the technical means to enable the cooperation without imposing new legal obligations.

7.2. The representative transaction

The walked example is a cross-border retail Open Finance transaction with embedded data exchange — the use case that combines both the Aperta-pattern data layer and the Agorá-pattern settlement layer at the consumer interface where adoption decisions are most directly made. A French consumer, resident in Lyon, purchases a high-value item — a refurbished musical instrument — from a small business in Manchester, United Kingdom. The consumer pays in euro from her primary current account at a French commercial bank, using an Open Banking payment initiation flow embedded in the merchant's checkout. The merchant receives sterling into its business account at a United Kingdom commercial bank. The transaction is medium-value (approximately three thousand euros), above the threshold at which authorised push payment fraud risk and Travel Rule obligations become material, and below the threshold at which it would be classified as a wholesale flow. It is the kind of transaction that today routes through a chain of correspondent intermediaries, takes one to three business days, and exposes both parties to fraud, fee opacity and dispute friction. It is also the kind of transaction whose aggregate volume, across European retail cross-border flows, drives the fraud loss figures cited in Section 6. The transaction proceeds through nine numbered steps, summarised in Figure 5.

Step one — initiation and consent verification. The consumer selects "pay by bank" at the merchant's checkout, authenticates into her French bank's Open Banking authorisation flow under PSD3 strong customer authentication requirements, and authorises both the payment initiation and a limited data exchange envelope that allows the merchant to receive real-time payment status notification. The consent coordination agent verifies that the consent envelope is complete and unambiguous, validates the lawful basis under GDPR for the cross-border data transfer to the United Kingdom under the adequacy decision, records the consent in the consumer's dashboard exposed by her bank, and confirms that the consent is current and unrevoked. The consent is bounded by purpose, value and time: this single transaction, this counterparty, this thirty-minute window.

Step two — payload validation and structured semantics. The data portability agent validates the payment instruction payload against the ISO 20022 Cross-Border Payments and Reporting Plus profile. The Legal Entity Identifier for the merchant is resolved against the Global Legal Entity Identifier Foundation registry; the consumer is identified by hashed national identifier under the

institution's privacy-preserving identity protocol. Confirmation of Payee runs against the United Kingdom beneficiary bank's verification service, with the result — name match confirmed — routed back to the consumer's bank interface before the transaction advances. This step alone forecloses the substantial share of authorised push payment fraud cases that rely on impersonation of legitimate beneficiaries, which represent the largest category of consumer-borne loss in the United Kingdom data cited in Section 6.

Step three — parallel compliance processing. The AML and sanctions coordination agent triggers screening simultaneously at the originator bank and the beneficiary bank. The French bank evaluates the consumer's transaction history, behavioural baseline and any sanctions exposure under European Union and French national lists. The United Kingdom bank evaluates the merchant against United Kingdom sanctions lists, the consolidated United Nations list and its own commercial risk parameters. Both screenings execute in parallel under each institution's own perimeter and own legal basis. Match metadata, false-positive feedback signals and confirmed typology indicators flow into the institutional memory tier under the consent and legal-basis gating described in Section 5.4. No personal or commercially sensitive data leaves either institution's perimeter; only the orchestration metadata — confirmation of completion, aggregate risk score band, hash of any match for cross-reference — crosses the corridor.

Step four — fraud detection. The fraud detection agents at the originator and beneficiary banks evaluate the transaction against their respective persistent typology memories. The originator-side agent compares the transaction to the consumer's behavioural baseline: amount range, beneficiary geography, device fingerprint and contextual signals. The beneficiary-side agent compares the inbound transaction to typology indicators for the merchant's segment and prior fraud patterns affecting United Kingdom small business beneficiaries of cross-border European Open Banking payments. In the representative case neither agent's bounded-autonomy policy identifies a signal exceeding escalation threshold and the transaction advances. Critically, if either agent had identified an authorised push payment scam pattern — for example, if the merchant's account had been recently flagged across the corridor's institutional memory — the bounded-autonomy policy would have triggered a real-time intervention notification to the consumer through her bank's interface, with explicit warning language and confirmation prompt, before the transaction advanced. This is the structural mechanism by which the fabric addresses authorised push payment fraud at the consumer-interface level, not just at the post-event reimbursement level.

Step five — pre-settlement validation. The settlement orchestration agent assembles the pre-validated payload for the Agorá-pattern settlement leg, locks the euro liquidity in the consumer's bank's tokenised commercial bank deposit position on the unifying ledger, performs the foreign exchange rate calculation through the corridor's reference rate mechanism, and prepares the atomic settlement instruction. The parallel compliance results from step three are aggregated and confirmed complete; settlement is conditional on this confirmation by the architecture's invariant.

Step six — atomic settlement. The Agorá-pattern atomic settlement executes across the unifying ledger and the two jurisdictional ledgers in a single transaction. The tokenised commercial bank deposit debit at the French bank, the credit at the United Kingdom bank in sterling, the central bank reserve movement between the Banque de France ledger and the Bank of England ledger to settle the foreign exchange leg, and any liquidity-conserving netting arrangement managed by the liquidity coordination agent — all occur or none occurs. Settlement risk is eliminated by construction [5]. Settlement finality is determined under the legal frameworks of the respective

jurisdictions, with the legal-finality analysis from Project Agorá providing the contractual baseline.

Step seven — post-settlement reconciliation and consumer notification. The settlement orchestration agent generates the reconciliation payload, updates both banks' internal records via their existing reconciliation interfaces, and produces the audit log entry for the supervisory observability layer. The consumer receives a real-time confirmation through her bank's interface showing the sterling amount credited to the merchant, the exchange rate applied, the total fees deducted with their itemisation under the PSR fee transparency obligation, and the transaction reference for any subsequent dispute.

Step eight — embedded data exchange. The data portability agent executes the consent-authorised data exchange leg: the merchant's checkout receives confirmation of payment with the transaction reference, the merchant's order management system receives the structured remittance information enabling automatic dispatch of the instrument, and the consumer's bank-aggregated personal finance management application (where the consumer has authorised it) receives the categorised expenditure record. All data exchange operates under the consent envelope verified in step one, with no exchange exceeding the purpose, value and time limits the consumer authorised.

Step nine — observability, dispute readiness and learning. Every agent action across the transaction is logged with sufficient detail to support DORA incident reporting and AI Act human oversight obligations. The supervisory observability agent exposes the relevant view to each participating authority under its jurisdiction. The liability allocation agent records the participant chain and the fraud-screening results at each step, producing the audit trail that supports PSD3/PSR refund obligations should the transaction subsequently prove fraudulent. The fraud detection agents, the AML coordination agent and the data portability agent each update their respective episodic and institutional memory tiers with the transaction's outcome — a confirmed clean transaction is itself a learning signal that improves future detection by reducing false-positive rates over time.

Figure 5. Nine-Step Transaction Walkthrough — EU-UK Retail Open Finance Payment

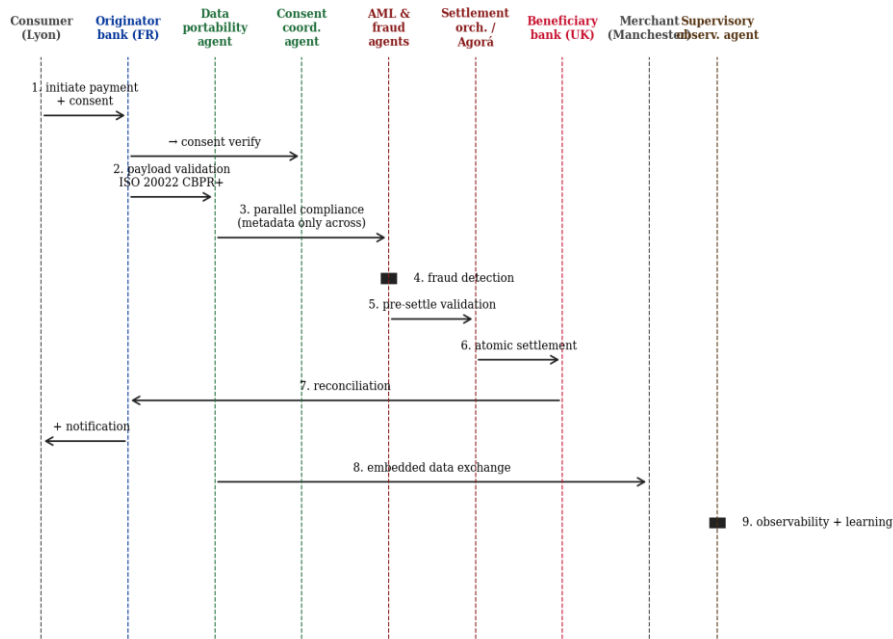


Figure 5. Nine-step transaction walkthrough for a representative retail Open Finance payment. Swimlanes show the consumer, the originator and beneficiary banks, the data portability, consent coordination, AML and fraud, settlement orchestration, and supervisory observability agents, the Agora settlement ledger, and the merchant. Numbered horizontal arrows trace the sequence of operations.

7.3. Gap-to-step closure mapping

The walked example closes specific gaps from Section 3 at specific steps in the transaction. This subsection makes the mapping explicit, demonstrating that the architecture's gap-closure claims are not theoretical but operational.

Gap one — coordinated AML, CFT, sanctions and anti-fraud approaches. Closed at step three, where the AML and sanctions coordination agent triggers parallel screening at both institutions with metadata-only cross-corridor information sharing. The mechanism delivers coordination without breaching jurisdictional data-sharing constraints, which gap nine separately requires.

Gap two — liquidity-saving mechanisms. Addressed at steps five and six, where the liquidity coordination agent participates in the atomic settlement payload assembly and the netting arrangement within the bounded-autonomy envelope.

Gap three — cybersecurity posture and operational resilience. Addressed continuously across all nine steps through the bounded-autonomy operation of every agent, the observability log produced at step nine, and the resilience and incident agent's monitoring of the transaction's progress against expected performance baselines.

Gap four — interoperability and compatibility requirements. Addressed at step two, where the ISO 20022 CBPR+ payload validation, the Legal Entity Identifier resolution and the Confirmation of Payee handshake demonstrate the corridor's compatibility with existing global payment infrastructure. The architecture sits alongside the existing rails, not above or against them.

Gap five — governance, rules and oversight. Addressed throughout by the governance agent's continuous operation, with the audit trail at step nine providing the documentation that supports both ex-post supervisory review and any subsequent rule-change consultation among participants.

Gap six — technical, operational and contractual definition of settlement finality. Addressed at step six, where the Agorá-pattern atomic settlement and its underlying legal-finality framework from the Project Agorá analysis provide the contractual baseline for finality across the two participating jurisdictions.

Gap seven — cross-border data portability without fragmentation. Addressed at step eight, where the data exchange operates under the Aperta-pattern interoperability framework and the consent envelope verified at step one, with no bilateral arrangements required between the institutions involved.

Gap eight — cybersecurity investment and cost asymmetry for smaller participants. Addressed by the federated deployment model exemplified throughout the transaction: the United Kingdom merchant's bank is not required to be a major institution; the architecture's cost surface is distributed such that mid-sized participants and smaller institutions can participate without bearing infrastructure costs that only large incumbents could absorb. The transition to a tokenised commercial bank deposit infrastructure by the merchant's bank does not require it to build the unifying ledger; it requires it to connect to one operated as shared infrastructure.

Gap nine — jurisdictional heterogeneity in risk aversion and data-sharing preferences. Addressed at steps one, three and eight, where the consent envelope, the parallel-then-metadata compliance processing and the consent-bounded data exchange respect the different legal bases applicable in France and the United Kingdom without requiring harmonisation of the underlying regimes.

Gap ten — continued refinement of Open Banking and Open Finance frameworks over time. Addressed by the learning loops in step nine, which feed transaction outcomes back into the institutional memory tiers and over time improve the fabric's detection and screening performance without requiring framework renegotiation.

Gap eleven — cross-jurisdictional learning between authorities. Addressed by the supervisory observability agent at step nine, which exposes parallel regulator-facing views to the European Banking Authority, the European Central Bank, the Prudential Regulation Authority and the Financial Conduct Authority — creating the technical substrate for cross-authority learning without imposing new legal cooperation obligations.

Gap twelve — participant adoption friction. Addressed structurally rather than at any single step, through the cost economics demonstrated by the transaction itself. The merchant's bank pays no participation premium beyond its ISO 20022 readiness and its connection to the Aperta data layer and the Agorá settlement leg; the participation cost is recovered through transaction fees on the flow itself.

Gap thirteen — regulatory acceptance heterogeneity for artificial intelligence components. Addressed throughout by the bounded-autonomy operation of every agent invoked. Each agent invocation in the nine steps operates under its EU AI Act risk classification (where applicable) and the parallel United Kingdom AI governance expectations, with documented purpose, scope, and human oversight provisions. No agent acts beyond its registered authority; every escalation is logged.

Figure 3 in Section 5.5 provides the comprehensive matrix view of these mappings; the present subsection traces them step-by-step against the walked transaction.

7.4. What the worked example demonstrates

The walked example is deliberately representative rather than exhaustive. It demonstrates four properties of the fabric that the prior architectural sections asserted but had not yet shown in operation. First, the architecture closes specific gaps from Section 3 at specific steps in a real transaction, as the explicit mapping in 7.3 establishes. Second, the cost discipline embedded in Section 5 manifests at every step: each agent invocation operates under a registered token budget, the tiered model routing means the great majority of steps invoke smaller models or rule engines, and the federated deployment means no single participant carries the full cost surface. The aggregate AI cost of the agent invocations across the nine steps is a small fraction of the per-transaction fee, leaving substantial margin for participant revenue. Third, the participant economics from Section 6 are realised at the transaction level: the originator bank captures its share of revenue from its part in the transaction, the beneficiary bank captures its share, the data exchange leg generates premium API revenue for the data portability agent's operator, the loss avoidance value of the coordinated fraud and AML detection accrues to all participants, and the merchant captures the operational benefit of real-time settlement against today's one-to-three day baseline. Fourth, the institutional protection primitives from Section 6.5 are operational in every step: liability allocation is predictable by construction, audit-grade observability is logged at every handoff, the dispute resolution path is defined in advance, no participant can change the rules mid-transaction, and the protocol layer is open and vendor-neutral.

The corridor as described is not yet production. The mechanisms that move it from prototype to production — sustainability, fundability, regulatory acceptance and adoption at scale — are the subject of Section 8.

8. DEPLOYMENT READINESS: SUSTAINABILITY, FUNDABILITY AND ADOPTION AT SCALE

The architecture described in Sections 5 through 7 is technically defensible and operationally specified. The remaining question is whether it can be deployed in practice and sustained at scale once deployed. This section addresses that question directly. It sets out eight structured mechanisms that together convert the architectural proposition into a deployable, fundable, regulator-acceptable and participant-adoptable infrastructure. It then develops an illustrative unit-economics model that quantifies the participant case for the EU–UK corridor under named assumptions. The section closes by addressing head-on the empirical record of artificial intelligence project abandonment in the enterprise sector, and explaining why the architecture proposed here is structurally protected against the failure modes that empirical record documents.

8.1. The eight deployment mechanisms

Phased rollout with reversibility checkpoints. The corridor advances through three explicit phases: pilot, controlled corridor expansion, and general availability. Each phase has defined exit criteria, success metrics and rollback provisions. Participants commit to one phase at a time and can withdraw at phase boundaries without stranded investment in the architecture's protocol layer. This pattern mirrors how the Bank for International Settlements Innovation Hub itself proceeds,

advancing from Project Helvetia to Project Jura to Project Agorá over multiple years and refining the design at each transition [5]. It also mirrors how Saudi Arabia, Bahrain and the United Kingdom each rolled out their Open Banking frameworks under phased regulatory perimeters. Participants budget for the phase they are in, not the full vision, which makes the budget conversation tractable.

Modular adoption through capability tiering. Participants opt into a subset of the agent population at lower entry cost. A bank may participate in the consent coordination agent and the AML coordination agent without committing to settlement participation; a payment service provider may participate in the data portability and fraud detection layers without operating tokenised commercial bank deposits. The lower entry threshold and smaller initial budget commitment preserve optionality and directly address gap eight from Section 3 — the cybersecurity investment cost asymmetry that disproportionately affects smaller participants. The corridor's value to any individual participant grows as the participant escalates through capability tiers; no participant is required to commit to the full architecture on day one.

Open standards, vendor neutrality, no rent-extraction at the protocol layer. The fabric's protocol layer is specified as an open reference architecture published under an industry-neutral license. No participant pays a per-seat or per-transaction license fee for the protocol itself. Commercial value sits at the application and services layer, where competition is permissible and expected. This is the SWIFT lesson, the FIX Protocol lesson, the ISO 20022 lesson: infrastructure standards adopted at scale are standards that no single party rent-extracts from. Without this commitment, no large institution will commit to participation because the lock-in risk at infrastructure level is unacceptable to its operational risk function.

Public-private cost-sharing through an industry utility model. Core infrastructure development cost is absorbed on a not-for-profit recovery basis through a permanent foundation or industry utility, analogous to The Clearing House in the United States, EBA Clearing in Europe, or the National Payments Corporation of India. Participants pay variable transaction fees tied to use, not fixed seat fees. The Bank for International Settlements and Institute of International Finance convening pattern that produced Project Agorá itself provides the reference for how public and private bodies share the convening and capital cost. The model has the additional virtue of aligning incentives: the utility succeeds only if participants succeed, since the utility's recovery is tied to transaction volume.

Regulatory sandboxes as the runway. Each participating jurisdiction's existing regulatory innovation environment becomes the live-fire testing environment for the corridor's pilot phase. The Financial Conduct Authority's regulatory sandbox, the European Banking Authority's innovation hub, the Bank of England's New Bank Start-up Unit, the Autorité de Contrôle Prudentiel et de Résolution's fintech-innovation interface and equivalent facilities in the European Union member states provide pre-existing legal, supervisory and operational infrastructure within which the pilot can run without requiring new legislation [2]. The sandbox approach addresses gap eleven from Section 3 — cross-jurisdictional authority learning — by embedding supervisor learning into the rollout from inception rather than as a downstream consultation.

Donor and multilateral co-financing for emerging-market corridor extensions. The eight mechanisms in this section apply uniformly across jurisdictions of varying development stages, but the financing model for integration cost differs. Major-market corridors (EU–UK, US–EU,

intra-GCC) can absorb integration cost through participant balance sheets. Emerging-market corridor extensions (EU–MENA, EU–Sub-Saharan Africa, GCC–East Africa, intra-Latin America) frequently cannot, and have historically depended on co-financing from multilateral development banks and bilateral donors including KfW, the European Bank for Reconstruction and Development, the World Bank, the African Development Bank, the Islamic Development Bank, the Agence Française de Développement and European Union Delegations [31]. Naming this mechanism explicitly opens the architecture to a much wider adoption surface than Aperta and Agorá contemplate. The EU–UK corridor itself requires no such support; its inclusion in this paper as the worked example serves to demonstrate the architecture, not to constrain its application.

Quantifiable participant economics with named assumptions. The participation case must be defensible at the board level, which requires quantified unit economics with explicit assumptions and sensitivity ranges. Section 8.2 develops the illustrative model for the EU–UK corridor.

Bridge to existing standards rather than displacement. The architecture sits alongside SWIFT, ISO 20022, FAPI 2.0, existing PSD2 and PSD3 obligations, the Cross-Border Payments and Reporting Plus profile, and existing card scheme rails. No participant has to retire an existing system to participate, which is the Project Agorá design choice — the platform operates alongside existing infrastructures — and which is the single most important adoption lubricant. It also forecloses the categorical objection that the architecture is asking participants to displace SWIFT or any other incumbent. It is not.

8.2. Illustrative unit economics for the EU–UK corridor

The illustrative model is anchored on a mid-sized European bank processing twenty thousand cross-border retail and small-business transactions per month into the EU–UK corridor, with an average transaction value of two thousand five hundred euros. The figures below are explicitly illustrative, assumption-anchored, and intended to demonstrate the structural cost case rather than to serve as a production benchmark. Each assumption is named.

Assumption set A — current state baseline. The institution absorbs an estimated fraud loss of fifteen to thirty basis points on its cross-border flow, consistent with the European Economic Area first-half 2023 baseline cited in Section 6 [28]. On twenty thousand transactions monthly at two thousand five hundred euros average value, this represents seven hundred fifty thousand to one million five hundred thousand euros in annual fraud loss attributable to this corridor flow. Operational cost of compliance and reconciliation against this flow, at industry-typical levels of fifty to one hundred basis points, adds a further two million five hundred thousand to six million euros annually.

Assumption set B — unilateral unbounded AI deployment. The institution invests in an in-house AI capability targeting fraud, AML and reconciliation improvements. Industry data from the FinOps Foundation, Mavvrik, Benchmarkit and LeanOps audits cited in Section 5 places the all-in cost of such deployments in the range of one to seven million United States dollars annually, with cost predictability poor — only fifteen percent of enterprises forecast within ten percent of actual [23] — and an abandonment probability above forty percent within twenty-four months [14][15]. Even where deployment succeeds, the value capture is institution-internal: signals cannot be shared with counterparties without further legal and architectural investment.

Assumption set C — corridor participation under the proposed fabric. The institution participates at a defined capability tier in the corridor. Per-transaction participation fee under the public-private utility model is assumed at five to fifteen basis points of transaction value, tied directly to use. Annual fee burden on the twenty thousand monthly transactions therefore falls in the range of three hundred thousand to nine hundred thousand euros. AI invocation cost is bounded by construction: each agent invocation in the nine-step transaction flow described in Section 7 operates under a registered token budget; tiered model routing means the great majority of invocations execute on smaller models or rules engines; the LeanOps audit benchmark of fifty-five to seventy-five percent cost reduction against unbounded baselines is applicable [24]. Fraud loss is reduced through coordinated detection. Assuming the coordinated fraud agent achieves a thirty to fifty percent reduction in fraud rate against the institutional baseline — consistent with the documented forty percent reduction achieved by mature machine learning fraud prevention deployments [29] — the residual fraud loss falls to three hundred seventy-five thousand to one million fifty thousand euros annually. Compliance and reconciliation cost is reduced through the parallel-then-coordinated AML pattern and the Agorá-pattern atomic settlement leg, conservatively assumed at twenty to forty percent reduction against the institutional baseline. Total annual cost under assumption set C falls in the range of one million five hundred thousand to four million euros, against four million two hundred fifty thousand to nine million euros under set A and two million to nine million seven hundred fifty thousand euros under set B inclusive of unbounded AI build cost.

The illustrative comparison shows the participant case is structurally defensible. Set C is consistently lower than set A across the assumption range. Set C compares favourably to set B at the low end of B's cost range, and is substantially lower than B at B's high end. Set C also has the structural advantage of cost predictability that B lacks: participation fees tied to transaction volume are linear and forecastable, where unbounded AI build cost is documented to be highly variable. The board paper writes itself, provided the architecture is honest about its assumptions, which this paper is. Figure 6 summarises the comparison.

Figure 6. Illustrative Unit Economics — EU-UK Corridor Participation

Mid-sized European bank · 20,000 cross-border transactions / month · €2,500 average value

Cost component	Set A: Current baseline	Set B: Unilateral AI	Set C: Corridor participation
Fraud loss absorbed	€0.75M - €1.5M	€0.45M - €0.9M	€0.375M - €1.05M
Compliance & recon. cost	€2.5M - €6.0M	€1.5M - €3.6M	€1.0M - €2.4M
AI build & operations	—	€1.0M - €7.0M	— (bundled in fees)
Participation fee	—	—	€0.3M - €0.9M
Cost predictability	High (status quo)	Low: $\pm 100\%$ + var.	High: linear, fee-tied
Abandonment risk in 24m	N/A	>40% (Gartner)	Negligible (no build)
Total annual cost range	€4.25M - €9.0M	€2.0M - €9.75M	€1.5M - €4.0M

Illustrative figures under named assumptions. Set C corridor participation is structurally lower than Set A across the range and compares favourably to Set B inclusive of unbounded AI build cost, with the additional structural advantage of cost predictability.

Figure 6. Illustrative unit economics for a mid-sized European bank participating in the EU–UK corridor under three assumption sets: current baseline, unilateral unbounded AI deployment, and corridor participation. Figures are illustrative under named assumptions and intended to demonstrate the structural cost case.

8.3. Why this architecture is structurally protected against the AI abandonment pattern

The forty-two percent enterprise AI abandonment rate documented in Section 5 [14] is not a curse on agentic AI as such; it is the consequence of three specific architectural failures, each of which the proposed fabric forecloses by construction.

The first failure is unbounded scope. Enterprises that bought capability rather than governance defaulted to using their AI for everything, with no role taxonomy and no termination conditions. The fabric's role-taxonomised ten-agent population, each with a registered mandate and an explicit autonomy budget, cannot suffer this failure: every agent has a defined remit and a documented termination condition.

The second failure is token-maxing — routing every task to the most capable and expensive model. The fabric's tiered model routing as a protocol-layer requirement, not a participant-side optimisation, prevents this by construction. Routine work cannot be routed to frontier models because the routing policy does not permit it. Frontier capability is escalated to only when bounded-autonomy policy requires it.

The third failure is invisible cost. Enterprises that could not measure their AI spend could not control it. The fabric's observability layer logs every agent invocation with token consumption attributed to participant, agent role and transaction. Every participant has the data to forecast and govern its own cost surface, and the cost visibility itself is enforced by the protocol, not optional.

The architecture proposed in this paper is therefore not a hopeful candidate for the eleven percent of AI deployments that succeed [14]. It is an architecture designed against the empirically documented failure modes of the forty-two percent. The deployment-readiness case rests on this distinction.

9. RISKS, OPEN QUESTIONS AND DISCUSSION

The architecture is not without risks. Honest treatment of those risks is part of the deployment-readiness case the prior section makes. This section sets out the principal risks and the residual open questions, and indicates the design responses each requires.

9.1. Model governance under multiple regulators

The agentic fabric operates under the European Union Artificial Intelligence Act, the United Kingdom's pro-innovation approach to AI regulation, sectoral guidance from the European Banking Authority, the Bank of England Prudential Regulation Authority, the Financial Conduct Authority and equivalent supervisors elsewhere [12][13]. Model documentation, validation, monitoring and retirement requirements differ across regimes. The risk is that participating institutions face inconsistent model governance obligations across the corridor, with the fabric's agents subject to different expectations depending on which institution operates them. The design response is to adopt the highest-bar regime as the corridor's baseline — in practice, the EU AI Act's high-risk classification requirements where applicable — and to operate the entire fabric to that standard regardless of any individual participant's jurisdictional minimum. This trades higher compliance cost against the elimination of cross-jurisdictional governance arbitrage and reduces supervisory friction at audit.

9.2. Adversarial adaptation in cross-border fraud campaigns

Cross-border fraud campaigns adapt continuously to detection mechanisms. The Nasdaq Verafin 2026 report documents that bad actors are themselves leveraging AI to scale fraud volumes, with the 9.2 percent year-over-year increase in fraud losses attributable largely to this adversarial trend [26]. The risk is that the fabric's fraud detection agents fall behind adversarial adaptation, particularly in the early phase before the institutional memory tier has accumulated sufficient typology indicators. The design response is the continuous learning loop embedded in Section 5.4, with the institutional memory tier propagating confirmed typology indicators across the corridor under documented legal basis. The persistence of memory across participants is the architectural mechanism by which the fabric maintains parity with adversarial adaptation; the bounded-autonomy operation of every learning loop prevents the parallel risk of the fabric itself developing adversarial patterns.

9.3. Sovereignty under operational resilience regimes

The Digital Operational Resilience Act imposes binding requirements on third-party ICT service providers including any party providing ICT services to a regulated financial entity in the European Union [20]. The corridor's shared infrastructure, the unifying ledger and the orchestration agents will likely fall within DORA scope. The risk is that DORA classification of corridor infrastructure as a critical ICT third-party imposes concentration risk constraints that constrain corridor design or scaling. The design response is to architect the corridor such that no single ICT provider operates more than a defined share of the infrastructure, with provider diversity built into the technical specification. The federated deployment model from Section 5.1 already supports this; explicit DORA concentration risk analysis must be part of the pilot phase exit criteria.

9.4. Privacy-enhancing technologies and the metadata boundary

Section 5 specifies that compliance coordination across institutions shares only metadata — confirmed typology indicators, match metadata, false-positive feedback signals — not underlying personal or commercially sensitive data. The risk is that the metadata boundary, while compliant with GDPR and UK GDPR on its face, leaves residual privacy attack surfaces particularly where metadata aggregation across institutions becomes re-identifying. The design response is the integration of privacy-enhancing technologies — secure multi-party computation, federated learning with differential privacy guarantees, and homomorphic operations on sensitive fields — into the institutional memory tier's cross-participant aggregation. The fabric's design accommodates these technologies; their selection and parameterisation is a downstream engineering decision that the pilot phase must close.

9.5. Regulatory acceptance heterogeneity beyond the EU–UK pairing

The worked example sits between two jurisdictions with mature, broadly aligned regulatory regimes. Extension of the corridor pattern to other pairings — EU–MENA, GCC–South Asia, intra-Africa — encounters jurisdictions with less developed or less harmonised regulatory baselines. The risk is that the architecture's regulatory acceptance is jurisdiction-specific in ways that do not generalise. The design response is the modular regulatory mapping in Section 4: the eight regime types named there are not exhaustive, and corridor extensions to jurisdictions with different regimes require explicit re-mapping of the regulatory scaffolding before agent

deployment. The architecture is robust to the mapping exercise; what cannot be promised is that every jurisdiction will accept the mapping outcome without bilateral negotiation.

9.6. Political-economy risk: central bank participation under policy shifts

The architecture as worked through the EU–UK example benefits from a particular political-economy context. Both jurisdictions operate central banks with high operational independence, long-standing supervisory continuity, and institutional cultures that accept multi-year participation in cross-border infrastructure experiments. The Banque de France operating for the Eurosystem, the European Central Bank in its supervisory role, the Bank of England and the United Kingdom regulatory authorities together provide a baseline of policy stability that is not universal. The architecture proposed in this paper is presented as a universal model — a corridor pattern intended to apply between any pair or set of jurisdictions whose regulatory regimes can be mapped onto the eight-regime scaffolding of Section 4 — but the political-economy preconditions for sustained central bank participation differ substantially across the world's payment system jurisdictions.

The risk is concrete and asymmetric. In some jurisdictions, central bank participation in cross-border architectural commitments is subject to executive policy direction that can shift on a parliamentary or electoral cycle. A change of government, a change of central bank governor, a shift in monetary or capital account policy, or a geopolitical realignment may withdraw or qualify a central bank's commitment to operate a jurisdictional ledger in the corridor's settlement layer. The Agorá-pattern settlement leg, which depends on continuous central bank operation of the jurisdictional ledger holding tokenised central bank reserves, is structurally vulnerable to such withdrawal. The Aperta-pattern data interconnection layer is less directly vulnerable but can be affected by policy shifts that alter the legal basis for cross-border data flows.

The design response is threefold. First, the governance agent's rule-change provisions must include explicit protocols for participant withdrawal — orderly exit pathways that protect remaining participants and in-flight transactions, with pre-agreed timelines and dispute resolution mechanisms. The architecture cannot prevent political-economy shifts; it can only ensure that such shifts do not cascade into operational disruption for the participants that remain. Second, the federated deployment model of Section 5.1 limits the operational coupling between the corridor and any single central bank's participation: while the settlement leg depends on participating central banks operationally, the data interconnection layer and most of the agentic fabric continue functioning even where one settlement participant withdraws. Third, the modular adoption pattern of Section 8 means that corridors can include central banks at varying levels of commitment, with the deepest tier reserved for those whose operational mandate is institutionally stable and shallower tiers available to participants whose commitment may need to be revisited under policy review.

The architecture's universality, in other words, depends on the architectural acceptance that not all participating central banks will look like the Banque de France or the Bank of England. Corridors involving jurisdictions where central bank participation may be qualified by policy shift must build that contingency into the corridor's design from the outset, rather than treating the EU–UK political-economy baseline as the default. The eight deployment mechanisms in Section 8 are deliberately framed at a level of generality that accommodates this variance, but the explicit acknowledgment of the political-economy risk is owed.

9.7. The participant acquisition cold-start problem

Sections 6 and 8 develop the participant economics and the acquisition path, but they do not eliminate the cold-start problem inherent in network-effect infrastructure. The first ten participants face an adoption case weaker than the case the hundredth participant will face. The risk is that the corridor stalls below critical mass and never realises the network value the architecture promises. The design response combines the public-private convening authority of the BIS–IIF model with the capability tiering that lowers entry threshold for the early adopters. Realistic deployment also benefits from anchor commitments by one or two systemically important participants in each jurisdiction whose presence catalyses broader participation; securing these commitments is a deployment activity rather than an architectural feature.

9.8. Settlement finality residual questions

Project Agorá's legal analysis confirmed settlement finality is achievable across all seven participating jurisdictions [5], but defined the technical, operational and contractual requirements as a separate workstream. The risk is that settlement finality in production deployment encounters edge cases — cross-default scenarios, insolvency intersections, corner cases in the legal interpretation of atomic settlement under specific national insolvency regimes — that the Agorá legal analysis did not fully characterise. The design response is the explicit incorporation of legal-finality dispute resolution into the governance agent's workflow, with pre-agreed arbitration provisions and clear jurisdictional escalation paths.

9.9. Open questions appropriate to subsequent research

Three substantive questions remain outside the scope of the present paper and indicate the research agenda the architecture opens.

First, the empirical performance of the architecture against the gap-closure claims of Sections 3 and 7 requires controlled pilot data, which can only be produced by a sandbox deployment. The paper specifies the architecture; the validation is a deployment activity.

Second, the optimal governance model for the public-private utility — board composition, voting rights, fee-setting authority, change management — remains a design space the architecture does not foreclose. The Bank for International Settlements convening pattern is one reference; The Clearing House and EBA Clearing are others; the National Payments Corporation of India is a fourth. The choice among them is jurisdiction- and corridor-specific.

Third, the extension path from the EU–UK pairing to multi-corridor topologies — corridors that share participants, that interconnect at common hubs, that operate under partially overlapping regulatory perimeters — requires explicit architectural work that this paper does not undertake. The fabric's design accommodates such extension; the engineering of it is the next research increment.

10. CONCLUSION

The architecture proposed in this paper extends a prior thesis on agentic AI in national-scale fraud detection [7] to the cross-border Open Finance layer where the Bank for International Settlements has, through Projects Aperta and Agorá, demonstrated that the data and value layers of cross-border financial infrastructure can be re-architected on shared, programmable substrates [4][5].

Both BIS projects achieve foundational proof. Both also name explicitly the work that remains. This paper consolidated those named gaps — six from the Project Agorá final report, five from the Project Aperta analytical work and the BIS Open Finance evidence base — into a thirteen-gap framework that also includes two structural limitations the present paper identifies in participant adoption friction and regulatory acceptance heterogeneity for artificial intelligence components.

The paper proposes a layered, role-taxonomised, cost-bounded agentic AI fabric that closes the thirteen gaps through ten agent roles organised into compliance and integrity, data and value orchestration, and supervisory and resilience families. The architecture is anchored in eight regulatory regimes — FATF, PSD3 and the Payment Services Regulation, ISO 20022 CBPR+, GDPR and UK GDPR, MiCA, DORA, the EU AI Act and parallel UK AI governance — as design constraints, not as downstream compliance work. The participant economics framework converts the architectural proposition into a structured cost-benefit comparison against the empirically documented alternatives: continued absorption of rising fraud loss rates or unilateral unbounded AI deployment with a forty-two percent abandonment rate [14]. The institutional protection layer ports the mature consumer protection literature up to the participant interface, providing predictable liability allocation, audit-grade observability, fair access to redress, protection against unilateral rule changes, and protection against vendor lock-in.

The architecture was walked through an EU–UK Open Finance corridor with an Agorá-pattern settlement leg, demonstrated through a representative retail transaction across nine numbered steps, with each step mapped explicitly to the gaps it closes. The deployment-readiness case rests on eight structured mechanisms — phased rollout, modular adoption, open standards, public-private cost-sharing, regulatory sandboxes, donor co-financing for emerging-market extensions, quantified participant economics, and bridge-to-existing-standards architecture — supported by an illustrative unit-economics model that demonstrates participation is incentive-compatible for a representative mid-sized European bank under named assumptions.

The forty-two percent enterprise AI abandonment rate documented in the contemporary record is not a curse on agentic AI as such; it is the consequence of three specific architectural failures — unbounded scope, token-maxing and invisible cost — each of which the proposed fabric forecloses by construction. The architecture is therefore not a hopeful candidate for the eleven percent of AI deployments that succeed but is designed against the empirically documented failure modes of the forty-two percent.

Two papers now exist on the agentic AI thesis. The first treated national-scale fraud detection infrastructure and demonstrated the architecture's value in a single jurisdiction under a single regulator [7]. The second, this paper, treats cross-border Open Finance and demonstrates the architecture's value across jurisdictions under multiple regulators. Together they sketch a research program: agentic AI as the orchestration substrate for the next generation of financial infrastructure, bounded by autonomy policy, observable by construction, economically defensible by design, and protective of every participant — consumer, institution and supervisor alike — by intentional architectural commitment.

The next increment of work is empirical. The architecture is sufficiently specified for a sandbox deployment in any participating jurisdiction's regulatory innovation environment. The Bank for International Settlements has shown that the substrate of cross-border financial infrastructure is

now programmable. This paper proposes that the workflow above that substrate be agentic. The case for both — programmable substrate and agentic workflow — is now made. The work that remains is to deploy.

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TABLE OF ABBREVIATIONS

AISP	Account Information Service Provider
AML	Anti-Money Laundering
API	Application Programming Interface
APP	Authorised Push Payment
BaaS	Banking-as-a-Service
BIS	Bank for International Settlements
CBDC	Central Bank Digital Currency
CBPR+	Cross-Border Payments and Reporting Plus
CDFI	Community Development Financial Institution
CFO	Chief Financial Officer
CFT	Countering the Financing of Terrorism
DORA	Digital Operational Resilience Act
EBA	European Banking Authority
ECB	European Central Bank
EMI	Electronic Money Institution
EU	European Union
FATF	Financial Action Task Force
FCA	Financial Conduct Authority
FI	Financial Institution
GDPR	General Data Protection Regulation
HLD	High-Level Design
IIF	Institute of International Finance

ISO 20022	International Organization for Standardization standard 20022 (financial messaging)
LEI	Legal Entity Identifier
MFI	Microfinance Institution
MiCA	Markets in Crypto-Assets Regulation
NBFI	Non-Bank Financial Institution
OECD	Organisation for Economic Co-operation and Development
PI	Payment Institution
PISP	Payment Initiation Service Provider
PRA	Prudential Regulation Authority
PSD2 / PSD3	Second / Third Payment Services Directive
PSP	Payment Service Provider
PSR	Payment Services Regulation
TPP	Third-Party Provider
UK GDPR	United Kingdom General Data Protection Regulation